

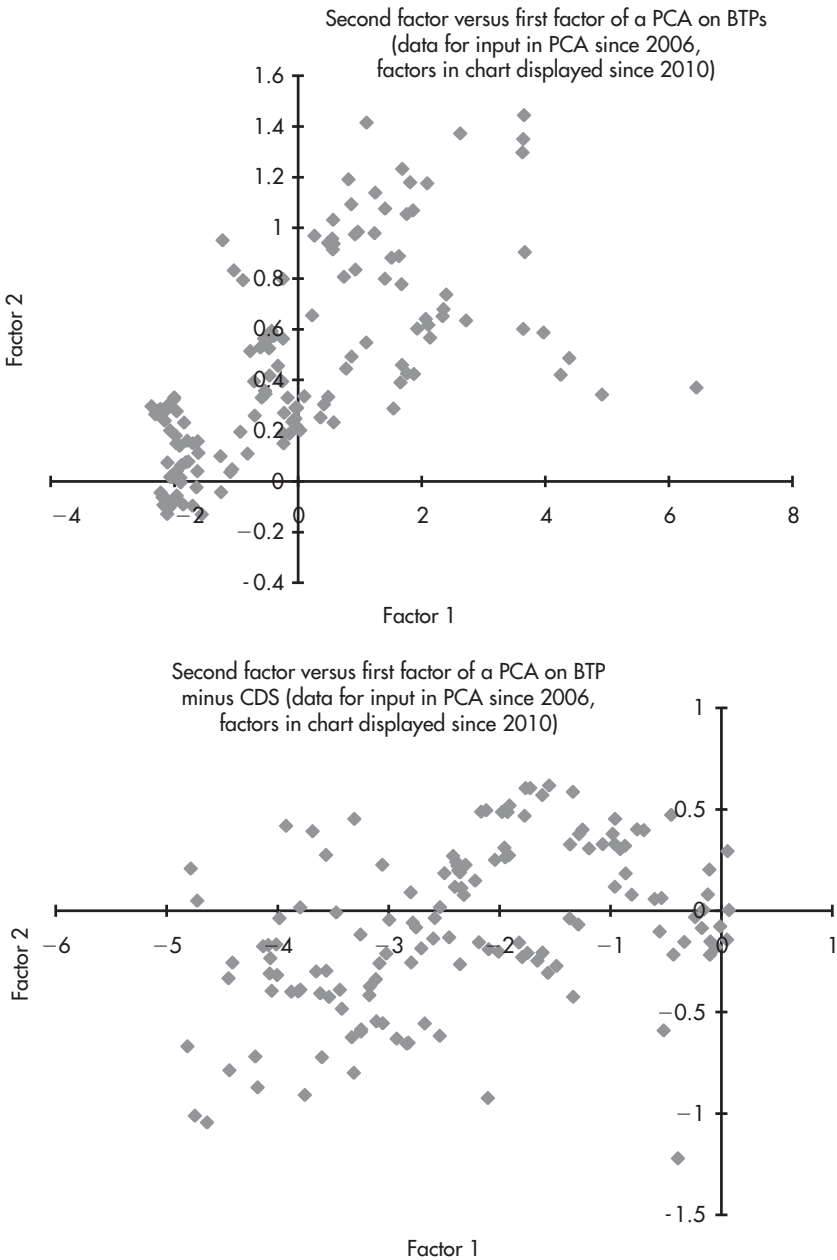


FIGURE 13.14 Correlation between factors 1 and 2 in the subperiod since 2010 of a PCA using BTP yields and of a PCA using risk-free BTP yields.

Sources: data – Authors, Bloomberg; chart – Authors.

Data period: 3 Jan 2010 to 26 Aug 2012, weekly data.

BTP curve trade based on pre-crisis PCA hedge ratios (without any credit impact in the data). Our actual P&L would be a function of an unforeseen impact of credit quality on both the hedge ratios and the yield curve dynamics.

- Likewise, the explanatory power of the first factor of a PCA on normal BTP data not only decreases generally in times of a shift in credit quality but its impact also becomes unevenly distributed over the yield curve. The explanation can again be found in the correlation between factors 1 and 2 following a shift in credit regimes. Now factor 2 contains part of factor 1, but the second eigenvector differentiates across the yield curve, causing a decrease of the explanatory power of factor 1 in a specific sector of the curve. This effect is summarized in Table 13.1, which compares the correlation of factor 1 of a normal and a risk-free PCA to different points on the yield curve.

TABLE 13.1 Correlation between Different Points on the Yield Curve and the First Factor of a PCA on BTP Yields and a PCA on Risk-Free BTP Yields

	BTP	BTP minus adjusted CDS
2Y	0.95	0.99
5Y	0.98	1.00
10Y	0.79	0.96

It turns out that the first factor of a PCA on normal BTP data has only limited explanatory power at the long end of the BTP curve. The PCA on risk-free data returns significantly better results; the explanatory power of the first factor over the BTP minus adjusted CDS curve over the five turbulent years during the euro crisis is comparable with a traditional PCA before the crisis and as evenly distributed over the yield curve.

- Factor 2 exhibits a significantly higher speed of mean reversion in the risk-free PCA. Intuitively, this can be understood by the fact that in the case of the PCA on normal BTP data some of the directional moves (with low speed of mean reversion) usually explained by factor 1 become part of factor 2 due to the shift in credit regimes. By contrast, factor 2 of the risk-free PCA does not need to bear any elements of factor 1 (reflected in a lower-scaled second eigenvalue in Figure 13.11) and shows therefore a speed of mean reversion typical for a factor 2 of bond yield curves in the absence of credit regime shifts (compare, e.g. Figure 13.13 with Figure 3.13).

- While CDS moves may sometimes on the surface appear to be irrational or erratic, the good PCA results suggest that they follow a rather stable, predictable pattern. This alleviates our concern regarding the use of risk-free data obtained by subtracting CDS quotes as inputs into statistical models.
- When (normal) bond yields approach their lower boundary (such as 0% or the ECB buying limit), their volatility tends to decrease, leading to unstable eigenvectors, particularly at the short end, particularly when accompanied by unusual monetary policy (see Figure 3.24 and Figure 3.25). This is a problem that can affect hedge ratios, for example. On the other hand, bond yields minus CDS are not bounded by zero and can therefore exhibit a higher volatility also in low-yield environments. This leads to a better stability of eigenvectors at the short end. Hence, using risk-free bond yields is an alternative to using shadow rates as discussed in Chapter 6 as input into a PCA in times of extremely low rates.

Turning from the theoretical benefits of risk-free bond yields to their practical exploitation, we observe that factor 2 of a PCA of risk-free BTP yields is away from its mean (Figure 13.15). This means that the BTP yield curve is too

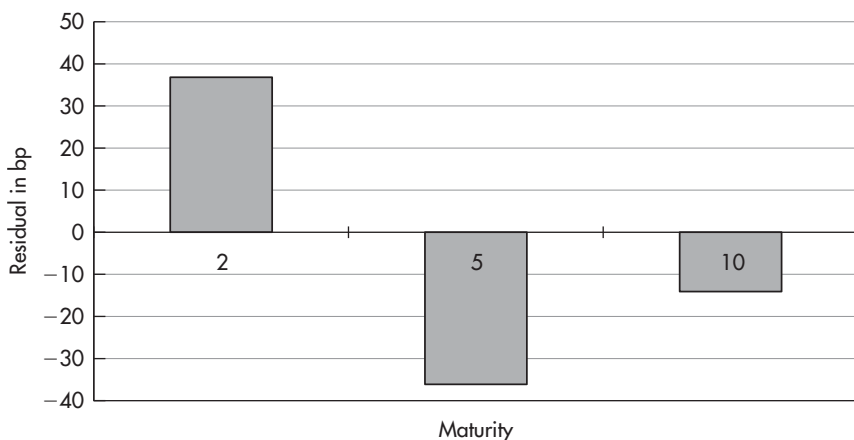


FIGURE 13.15 One-factor residuals of the PCA on risk-free BTP yields.

Sources: data – Authors, Bloomberg; chart – Authors.

Data period: 1 Jan 2006 to 26 Aug 2012, weekly data; “current” as of 26 Aug 2012. Note that the result is not materially affected by using shorter data periods (e.g. only post-crisis data). This is in line with the discussion above and a contrast to a PCA on normal BTP yields.

flat relative to the Italian CDS curve. In other words, steepening positions of the risk-free BTP yield curve (buying short end BTPs and short maturity Italy CDS versus selling long end BTPs and long maturity Italy CDS) are attractive. As always, one can choose the statistically most attractive expression for steepening positions on the BTP relative to the CDS curve by looking at the one-factor residuals.

One might therefore consider a 2Y-5Y steepener as a candidate for a trade idea. This would involve selling 5Y BTPs and 5Y CDS versus buying 2Y BTPs and 2Y CDS, with the hedge ratios being given by the first-factor sensitivities (resulting in a ratio of roughly 5:3 for the 2Y:5Y instruments).

Note that this trading opportunity does not show up in a PCA on the normal BTP curve. Hence, it arises from the *relationship* between the yield curve and the CDS curve, which is only visible through the risk-free PCA. The 2Y-5Y BTP curve is too flat *relative to the CDS curve*. The fact that few analysts consider the steepness of bond yield curves in relation to the CDS curve may help explain the unusually large profit potential of 73 bp for a relative value position on the yield curve.

PITFALLS

We can thus use the CDS to make statistical models work also in an environment of unstable credit risk. On the other hand, this introduces the particular features of the CDS, in case of using unadjusted CDS levels in particular the DO and FX components, and we recommend checking any trading strategy derived by statistical models on the CDS market for the following problems:

- **FX component.** If the CDS is denominated in a different currency than the government bond it protects, the holder of a long CDS position profits from the currency of the defaulting government weakening in case of default (and vice versa). For example, in the 2Y-5Y steepening trade discussed above, the notional weights (long 5:short 3) give us a net long position in the Italian CDS, thus in that case we seem to be on the profitable side of the FX exposure in case of default. Had the trading strategy resulted in a net short position of the CDS, on the other hand, we might have chosen to assess the profit potential and the expected holding horizon of the trade versus different estimates for the unquantifiable potential loss from the EUR weakening in case of an Italian default.
- **Delivery option.** Similarly, the net long or short position in the CDS decides about whether we are net long or short the DO in case of a default.

Again, in our example, we are net long the CDS, thus also the DO, and have passed this check. As a general rule, being net long, the CDS usually results in being on the safe side of potential issues surfacing in case of a default.

- **Repo.** What is the expected value of the bonds involved in the repo market? When approaching default, repo markets may well become dysfunctional, driving the actual costs away from the estimation. How would this affect the overall position?
- **In case of euro sovereign bonds being involved, the redenomination risk.** While the CDS is and remains USD-denominated, BTPs might be repaid in lira.¹² Thus, we face the problem that by construction of the bond versus CDS curve trades, if we are net long the CDS (which is the “safe side” concerning the FX component and DO), we are also net long the lira in case of BTP redenomination. In the example above, if Italy redenominates in lira and this does not fall under the default definition of the CDS, the steepener with a notional ratio of long 5:short 3 is exposed to a lira weakening. An investor assigning a significant probability to this risk may therefore want to use CDS treating redenomination as default, or, if these are too expensive, to refrain from applying these trading strategies to euro sovereign bond curves. For example, he could also enter a 2Y-10Y flattener on the risk-free JGB yield curve for an expected profit of 47 bp and then assess whether the exclusion of redenomination risk (or the saving on the cost for CDS covering it) was worth the loss of a 26 bp profit potential (and the bigger distance between the two points on the yield curve).

As a general note, given the high speed of mean reversion for most (good) relative value trades, chances are high that in reality they will not be subject to these problems. In other words, if the expected holding horizon of a statistical trade involving the CDS market is shorter than the time period over which issues arising from default can reasonably be expected to occur, then one can justify ignoring the potential problems and analytically treating trades involving CDS as if those problems did not exist. As mentioned in Chapter 1, one then has a useful but not necessarily theoretically rigorous statistical model for CDS trades. Still, we would like trades involving CDS to work also in the event of default and therefore advise checking for the above points, even when a high speed of mean reversion makes it unlikely that a real default will test our performance in this task.

¹²Whether redenomination counts as default has been discussed extensively and depends on the specifications of the individual CDS agreements.

CONCLUSION

- CDS can be priced either against other CDS through a statistical model or against the underlying bond as outlined above or against asset&basis swapped bonds through a financial model, covered in Chapter 16.
- The CDS curve over different maturities often exhibits a single-factor structure and thus permits little room for relative value trades.
- A PCA of the CDS of different issuers reveals the pricing mechanisms and potentially attractive relative value trading opportunities. For example, it segments the CDS of euro sovereigns into bailout and no-bailout countries, and the core group according to the size of the bond markets.
- A PCA of risk-free yield curves can alleviate some of the statistical deficiencies of a PCA of normal yield curves. Moreover, at times it detects significant relative value opportunities between the shape of the yield curve and the shape of the CDS curve.
- Since the DO and FX component of a CDS may be part of a curve trade on risk-free yields, every trading position involving the CDS should be assessed from the viewpoint of its performance in the event of default for a reasonable range of different values for the DO and the FX component.

Intra-Currency Basis Swaps

DEFINITION

An intra-currency basis swap (ICBS) is a derivative security in which each of the two parties agrees to pay the other a different floating rate, with both rates denominated in the same currency. For example, one party might agree to pay six-month (6M) LIBOR to the other party in exchange for being paid 3M LIBOR less a spread.

As described in Chapter 11, the transition away from LIBOR over the last decade has resulted in additional or alternative reference rates, such as unsecured overnight (O/N) rates and secured overnight financing rates (SOFR) in the US. Correspondingly, the ICBS definition has been extended: While it formerly referred to basis swaps between different LIBOR tenors only, it now also includes basis swaps between unsecured O/N rates and LIBOR (in those markets where LIBOR still exists) and between unsecured O/N rates and SOFR in the US.

In terms of Table 11.1, ICBS can thus be defined as basis swaps between two different reference rates in the same currency, i.e. in the same row, including basis swaps between two different terms of the same reference rate, i.e. LIBOR shown in the left column. Of particular importance are the following three ICBS:

- In Europe and Japan, ICBS between different tenors of EURIBOR and TIBOR.
- In Europe, ICBS between EURIBOR and ESTR. This basis swap plays an essential role in constructing a basis swap between SOFR and EURIBOR and hence in the following chapters. Reflecting its importance, while ICBS are usually over-the-counter (OTC) products, CME offers clearing for a standardized version of the EURIBOR-ESTR ICBS.
- In the US, ICBS between SOFR and Fed Funds. This ICBS covers the secured–unsecured basis and therefore has the vital function of providing a market price for that funding spread. In case of using swaps with SOFR as reference rate for asset swapping Treasuries and swaps with Fed Funds

as reference rate for asset swapping US corporate bonds (as discussed in Chapter 12), this ICBS connects both. Since there is also a market for spread future contracts between SOFR and Fed Fund futures with margin offset, this ICBS can be cheaply hedged and replicated via futures.

PRICING OF ICBS

Per definition above, pricing ICBS is a direct application of the models provided in Chapter 11 assessing the fair value of the spread between different reference rates in the same currency. Specifically:

- ICBS between different tenors of EURIBOR and TIBOR are driven by the term (risk), which includes a minor effect for short rate expectations, a minor effect due to interest rate compounding frequencies, and a catchall term that often is referred to as “risk premia.” The main driving force is the perception about the credit risk of (LIBOR panel) banks: The higher this risk, the more valuable the credit ‘refreshes’ explained in Chapter 11. In case of a banking crisis, the value of the option to switch counterparties after 3M rather than being stuck with one borrower for 6M increases and so does the 3M-6M LIBOR ICBS.
- ICBS between LIBOR and unsecured O/N rates can be priced in the same framework of term (risk). With O/N rates taking the advantage of credit ‘refreshes’ to the practical maximum, the ESTR-6M EURIBOR ICBS usually quotes above the 3M–6M EURIBOR ICBS (and the ESTR-3M EURIBOR ICBS), with the spread between both increasing as the perception of bank credit risk deteriorates.
- ICBS between SOFR and Fed Funds reflect the secured–unsecured basis, which is a function of the cost of capital (and the capital requirements). In a banking crisis, the cost of capital of banks is usually highly correlated to the perceived credit risk of banks. In these situations one can therefore expect the ICBS between different unsecured reference rates (e.g. ESTR and 3M EURIBOR) and the SOFR–Fed Fund ICBS to rise. Given the current situation of global reference rates, there is no ICBS between secured O/N rates (SOFR) and unsecured term rates (LIBOR). Thus, ICBS pricing models do not require the full repo–LIBOR model (including term risk), but can apply simpler versions, as outlined in Chapter 11.

To provide intuition about the impact of perceived credit risk of banks on the pricing of ICBS, Figure 14.1 shows the five-year (5Y) basis swap spread between 3M EURIBOR and 6M EURIBOR during the subprime and euro crises. The positive spread shown in Figure 14.1 depicts the spread attached



FIGURE 14.1 5Y basis swap spread between 3M EURIBOR and 6M EURIBOR.

Sources: data – Bloomberg; chart – Authors.

to the 3M rate in this case. For example, in June of 2010, one could agree to pay 6M EURIBOR for five years in exchange for receiving 3M EURIBOR + 18 basis points (bp) for five years.

A few points to note:

- The basis swap was almost always within 1 bp of zero prior to the onset of the subprime crisis in July of 2007.
- Subsequent to the onset of the crisis, the spread widened considerably, reflecting the perception that the risk of bank default had increased.

To further illustrate the relation between the level of the EUR 3M/6M basis swap and overall perceptions of crisis-related credit risks, we show the 5Y EURIBOR 3M/6M basis swap along with the euro exchange rate in Figure 14.2. In Figure 14.3, we show this ICBS along with the 5Y basis swap between 3M EURIBOR and 3M USD LIBOR.

Prior to the onset of the subprime crisis, there was no correlation between the euro and the ICBS, but after the onset of the crisis, the correlation between these two variables has been 0.68.

Prior to the onset of the subprime crisis in July of 2007, the two basis swaps were both very close to zero. Subsequent to the start of the crisis, the spreads have tended to widen and narrow together. Over the entire period,